

Thesis plan: Coherent states in numerical methods for fermionic particle-preserving quantum systems (Michal Horanský)

1 Introduction

1.1 Semi-classical methods in high-dimensional quantum mechanics

1.1.1 The problem of diagonalisation

1.1.2 The problem of dynamics

1.2 Coherent states throughout the 20th century

1.2.1 Schrödinger coherent states

1.2.2 Glauber coherent states

1.2.3 Gilmore-Perelomov coherent states

2 Constructing the coherent state manifold

2.1 Gilmore-Perelomov coherent states in general

2.1.1 Action group, stability group

2.1.2 Reference state operator and Hadamard's lemma

2.1.3 Topology of the coherent state manifold

2.1.4 Equations of motion

2.2 $SU(M)$ fermionic coherent states

2.2.1 Displacement operator

2.2.2 Decomposition into the full occupancy basis

2.2.3 Overlaps and normalisation

2.2.4 Fermionic operator sequence matrix elements

2.2.5 Overlap update formalism

2.3 Approximately-coherent particle-number-conserving states

2.3.1 Qubit operators

2.3.2 Overlaps with elementary symmetric polynomial

2.3.3 Fermionic operator sequence matrix elements

2.3.4 Topology of the ACPNCS manifold

3 Molecular electronic structure

3.1 System description

3.1.1 Non-relativistic electronic Hamiltonian

3.1.2 Spin- z -labelled subspaces

3.1.3 Symmetry-labelled subspaces

3.1.4 Potential energy surfaces

3.2 Ground state estimation with Monte Carlo

3.2.1 Low-excitation-guided sampling

3.2.2 Varying the sample basis size

3.2.3 Varying the Hilbert space dimension

3.2.4 The effects of spin and symmetry

3.2.5 Multireferential regimes

3.3 Potential energy surface extrapolation with coherent-state transfer

3.3.1 Sampling and decomposition strategy

3.3.2 Energy eigenstate reconstruction from transferred basis

3.3.3 Spin-protected crossings

3.3.4 Symmetry-protected crossings

3.3.5 Unprotected crossings

3.3.6 Multireferential regimes

3.4 Discussion

4 Appendix

A Counting constrained minor products

B Repeated commutator of the $SU(M)$ displacement operator and the $\hat{\phi}$ -operator